Ojasva Mishra

SWE & AI/ML Systems | Z Fellow | CMU SCS Researcher | Quantitative Research Analyst | Robotics & Autonomy
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EDUCATION

University of Florida, Gainesville, FL GPA 4.0
Bachelor of Science in Computer Engineering; dual major in Finance
Graduation Date: May 04, 2029
YC Startup School 2026 | Downingtown STEM Academy, IB Diploma | SAT 1590, National Merit Commended

EXPERIENCE

Drift (backed by Antler) November 2025 to Present
AI/Software Engineer (Founding Team) San Francisco, CA
- Built the **Gaussian splat pipeline**, my largest system: it turns ordinary photos of real spaces into walkable 3D worlds that robots are simulated in (ROS2, Gazebo).
- Built a **self-healing loop**: an error-triage service clusters user-reported failures, then a repair agent patches the code and reruns CI, so the codebase maintains itself.
- Built core infrastructure and internal tooling for a **robotics AI coding agent used by 1,000+ people** to build and debug simulations in ROS2, Gazebo, MuJoCo, and Isaac Sim (#5 Product of the Day at launch); Drift Desktop in Tauri (Rust) and React.
- Shipped **World Labs photo-to-3D scene generation** and a **129-robot catalog** license-checked against MuJoCo Menagerie; wrote a pexpect CLI eval harness graded by a vision model on Gazebo screenshots; traced a production Gazebo freeze to gz-transport UDP multicast on Ubuntu 24.04 and fixed it.

Twinly Incorporated December 2025 to Present
Co-Founder and CEO San Francisco, CA
- Build **digital clones** that learn a user's writing, speech, and computer behavior, then run their work in hundreds of parallel instances; **Z Fellows 10-builder cohort**.
- Engineered the **Identity Runtime**, the layer that lets a clone operate a real computer: it fuses OS accessibility APIs, browser DOM and app state, and multimodal vision into one execution graph with persistent memory; shipped on **macOS, Windows, and iOS** to **1,000+ active users, including paying customers**.

Carnegie Mellon University, Xu Lab July 2025 to Present
Research Intern Pittsburgh, PA
- **First author on three robotics and control papers** with Prof. Min Xu and Dr. Xiaolong Wu: optimize discrete-time PID gains so torque-limited joints stay stable when their motors saturate, and add conformal-prediction safety filters that bound how far a corrupted or attacked feedback signal can push the robot.
- Validated the controllers on a **Franka Emika Panda** arm; built multimodal SLAM, path planning, and constrained-control experiments.

AlgoGators Investment Fund September 2026 to Present
Quantitative Research Analyst Gainesville, FL
- Selected as **1 of 2 first-year students** for the quantitative research team at a **~$500K AUM** student-run fund; research and backtest systematic trading signals.
- Ran a **crack spread strategy** (crude vs. refined fuels), resizing as hurricane forecasts shifted: **64% win rate over 1,200 live trades**.

EV Bots, Inc. April 2025 to February 2026
AI Robotics Engineer Intern Chester Springs, PA
- Built the company's **first functional autonomous EV-charging robot prototype** end to end: mechanical design, electronics, perception, and closed-loop control that finds the car and connects to it; designed an autonomous drone-charging dock; contributed to **multiple provisional patents**.

PteroLabs July 2025 to January 2026
Software Engineering Intern (UAV Simulation and Autonomy) Remote, United States
- Built a **mission-grade Unreal Engine 5 UAV simulator** that runs real PX4, ArduPilot, and Betaflight flight code in the loop (SITL), with 6-DOF flight dynamics, FPV, ML mission planning, and object detection; adopted by the **University of Pennsylvania**.

iPipeline June 2025 to August 2025
Software Development Intern Exton, PA
- Shipped automated test frameworks, AI agents, and developer tooling adopted across engineering teams for regression testing (Python, Docker).

PROJECTS AND COMPETITIONS

Lucid Dreams | Gator Hack V, 48-hour build | Live | GitHub October 2026
- Turns a phone video or a few photos of someone into a **life-size, moving 3D person** in Meta Quest passthrough, from one browser link.
- 4DAnyone makes 48 synced views; gsplat fits ~290K Gaussians/frame on **12 NVIDIA B200s**; a WebXR client (three.js, Spark) renders the top 160K.

Déjà View | SwampHacks XII, **1st** AI/ML track and **1st** DeepSpace track | Live | GitHub | Video | Devpost October 2026
- Turns photos or video of any space, real or in-game, into a 3D world you can walk into on Meta Quest (WebXR), in a browser, or in Roblox.
- COLMAP camera solve, then a ~1M-Gaussian splat trained on UF HiPerGator in 11 min; VLM and depth gap fill; 240 Hz VR-Roblox cross-play physics.

Courtside | Gator Quant Hacks 2026, Systematic Trading track | Paper | GitHub | Video October 2026
- Causal computer-vision ball tracker that calls the point a median **162 ms before the bounce** on held-out table tennis, none wrong, at 119.9 fps.
- Latency-arbitrage study of **13,084 Polymarket ATP/WTA markets**: wallets trading within 3 s of a point were profitable after fees in all 11 out-of-sample months, as a pre-registered hypothesis predicted; Markov fair value, deflated Sharpe, block bootstrap. Paper trading only.

Volition | Brain-controlled exoskeleton concept | Live | GitHub October 2026
- Designed an exoskeleton for complete paraplegia driven by imagined movements read from a 16-channel EEG cap: parametric CAD (CadQuery), BLDC motors on 100:1 strain-wave gears, and a 1 kHz closed-loop simulation (design only) that stands, walks 8 steps, and sits from decoded intent (85% held-out accuracy).
- Recomputes each joint's torque limit every 1 ms as the min of 5 limits (peak, current, voltage, thermal, slew); anti-windup cut knee overshoot 1.69° to 0.38°.

Kinetix (NexHacks): AI physical-therapy coach; MediaPipe pose tracking scores joint angles, counts reps, and voice-coaches on Meta Quest (Next.js, LiveKit).

HIVE (YC Transpose, **1st** place): computer vision turns a physical workspace into task graphs that coordinate the people in it (PyTorch, React).

FUSE (Bitrig Hacks iPhone Duo Edition, Y Combinator): native iOS app; fold the iPhone Duo to fuse both screens into one AI result (Swift, RevenueCat).

GlassBoxAI (Startup in a Weekend Philly, **2nd** of 200+): Zoom/Google Meet overlay flagging AI-assisted interview answers via eye tracking and voice analysis.

Apps: PreventaAI (App Store, 5.0; SwiftUI, Firebase); Eyesight (LiDAR + on-device CoreML obstacle detection); QuickCitePro (Chrome extension, 5.0).

Technology Student Association, 2x National Finalist: 8 national events, captain for 6: SAFMS (AI/IoT carbon-credit app, on-chain), drone challenge (Betaflight FPV), VR visualization (Unity, Meta Quest), video game design (Unity, C#), systems control (Python, C++), engineering design, robotics.

VEX Robotics, 3x World Championship Competitor: captain, lead programmer, and driver; 7 seasons of C++/Python autonomy, PID, CAD, and sensors.

RESEARCH AND PUBLICATIONS

- **First author:** Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation. **IEEE T-RO**.
- **First author:** Bounding Detector-Accepted Feedback Authority in Torque-Limited Robot Control. **IEEE ICRA 2027**.
- **First author:** SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control Under Saturation. **IEEE/RSJ IROS 2026** (video).
- Introduction to Robotics and Programming, a 120-page open-access textbook indexed on SSRN.

LEADERSHIP

- **Dream Team Engineering, NeuroSpine Research Team:** with UF Neurosurgery, building a medical image segmentation pipeline for spine CT and MRI (Python, nnU-Net, TotalSegmentator, MONAI, SimpleITK) that automates Hounsfield-unit and VBQ bone-quality metrics for surgical planning.
- **American Rocketry Club, Founder** (60+ members): payload CAD, propulsion (MATLAB); **Youth Publications and Socioeconomic Forum, CTO** (15+ countries).

TECHNICAL SKILLS

Languages and frameworks: Python, C++, Rust, Java, C#, Swift, JavaScript, TypeScript, SQL, MATLAB, React, Next.js, Node.js, Tauri, SwiftUI, gRPC
Infrastructure: Linux, Git, Docker, CI/CD, AWS, GCP, Vercel, PostgreSQL, MongoDB, Firebase, Supabase, WebSockets, WebRTC, Slurm HPC
AI, ML, and quant: PyTorch, Gaussian splatting (gsplat, COLMAP), OpenCV, CoreML, VLMs, LLM agents, Claude Code, Codex, Cursor, backtesting, NumPy, pandas
Robotics and hardware: ROS2, Gazebo, MuJoCo, Isaac Sim, SLAM, PID control, PX4, ArduPilot, Arduino, CadQuery, Fusion 360, CNC, 3D printing, WebXR